

Surface Phenomena & Catalysis

Week 4: Selectivity Engineering

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Week	Chapter	Topic
1	Ch 1–2	Bridge Module + Computational Perspectives
2	Ch 3	Reactor Design & Catalyst Characterization
3	Ch 4	Transport Limitations in Porous Catalysts
4	Ch 5	Selectivity Engineering
5	Ch 6	Zeolites & Carbon Nanotubes

When multiple products can form, how do we maximise the desired one?

Today's Agenda

Part I: Definitions & Parallel Reactions

~30 min

Conversion, yield, selectivity, the S–Y–X triangle, parallel network solutions, temperature control

Part II: Consecutive Reactions & Design

~35 min

Selectivity window, optimal τ^* , PFR vs CSTR, temperature effects, industrial applications

Part III: Summary & Self-Study

~10 min

Key equations + NB5 self-study guide + looking ahead to Week 5

Lecture notes: Chapter 5 — Notebook: NB5

- 1 Part I: Definitions & Parallel Reactions
- 2 Part II: Consecutive Reactions & Design
- 3 Summary & Self-Study

Why Selectivity Matters

*In Weeks 2–3 we focused on converting A as completely as possible. But most catalytic processes produce **multiple products**.*

The new question:

- Maximising conversion alone is **not enough**
- We must also maximise the **desired product**
- Reactor type, τ , and T all affect product distribution

Industrial reality

Ethylene oxide on $\text{Ag}/\alpha\text{-Al}_2\text{O}_3$: only 8–12% per-pass conversion to avoid total combustion to CO_2 .

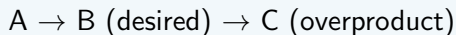
Two fundamental networks

Parallel:



Temperature is the lever.

Consecutive:



Residence time is the lever.

Today's goal

Learn to choose reactor conditions that maximise the **yield** of B.

The X-Y-S Triangle

Conversion:

$$X_A = \frac{C_{A,0} - C_A}{C_{A,0}}$$

Overall yield of B:

$$Y_B = \frac{C_B}{C_{A,0}}$$

Overall selectivity:

$$S_{B/A} = \frac{Y_B}{X_A}$$

The triangle relation

Any two quantities fix the third:

$$Y_B = S_{B/A} \times X_A$$

Quantity	Range
X_A	$[0, 1]$
Y_B	$[0, 1]$
$S_{B/A}$	$[0, 1]$

Physical meaning

$S_{B/A}$ = fraction of converted A that became the desired product B.

Instantaneous vs Overall Selectivity

Overall selectivity (integral, measured):

$$S_{B/C} = \frac{C_B}{C_C}$$

Ratio of accumulated products at reactor exit.

Instantaneous selectivity (point, local):

$$s_{B/C} = \frac{r_B}{r_C}$$

Ratio of formation rates at a given C_A .

When do they differ?

- Equal-order reactions: $s_{B/C}$ is constant
 $\Rightarrow S_{B/C} = s_{B/C}$
- Different-order reactions: $s_{B/C}$ depends on $C_A \Rightarrow S_{B/C}$ depends on reactor type and X_A

Key insight

Instantaneous selectivity tells us what **local conditions** favour B. Then we choose the reactor that provides those conditions.

The Selectivity–Conversion Tradeoff

The dilemma:

- High $X_A \Rightarrow$ more product, but selectivity often **decreases**
- High $S_{B/A}$ often requires **low per-pass conversion**

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Industrial solution: recycle.

- Operate at low X_A per pass (high $S_{B/A}$)
- Separate product, recycle unreacted A
- Effective $Y_{B,\text{overall}} \approx S_{B/A} \times X_{\text{overall}}$

Industrial examples

Process	X (per pass)	S (w/ recycle)
Ethylene oxide (C_2H_4)	8–12%	>95%
Methanol (CO_2)	~25%	>99%

Process chemist's challenge

Find the operating point that maximises **profit**:

(value of B) – (cost of separation + recycle)

Concept Check

1. A reactor is fed 100 mol/h of pure A. At the exit: 30 mol/h A, 50 mol/h B, 20 mol/h C. Calculate X_A , Y_B , and $S_{B/A}$.

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 $X_A = 0.70$, $Y_B = 0.50$, $S_{B/A} = 0.50/0.70 = 0.714$
2. If $X_A = 0.60$ and $Y_B = 0.42$, what is Y_C ?

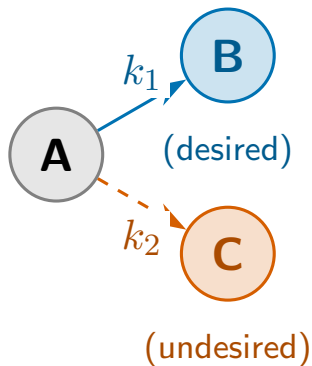
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 $Y_C = X_A - Y_B = 0.60 - 0.42 = 0.18$ (by stoichiometry)
3. A process has $S_{B/A} = 0.95$ at $X_A = 0.10$ but $S_{B/A} = 0.60$ at $X_A = 0.80$. Which operating point would you choose, and why?

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3. A process has $S_{B/A} = 0.95$ at $X_A = 0.10$ but $S_{B/A} = 0.60$ at $X_A = 0.80$. Which operating point would you choose, and why? **Low X_A + recycle:** $Y_B \approx 0.95 \times 0.10 = 0.095$ per pass, but near-quantitative overall with recycle

Parallel Reactions: Network & Rates



First-order kinetics:

$$r_1 = k_1 C_A \quad (\text{desired})$$

$$r_2 = k_2 C_A \quad (\text{undesired})$$

Net rates

$$R_A = -(k_1 + k_2) C_A$$

$$R_B = +k_1 C_A$$

$$R_C = +k_2 C_A$$

Observation

Both rates have the **same** dependence on C_A . Changing C_A (or τ) affects both equally.

Parallel Reactions: PFR & CSTR Solutions

PFR (integrate $dC_i/d\tau = R_i$):

$$C_A = C_{A,0} e^{-(k_1+k_2)\tau}$$

$$C_B = C_{A,0} \frac{k_1}{k_1 + k_2} [1 - e^{-(k_1+k_2)\tau}]$$

$$C_C = C_{A,0} \frac{k_2}{k_1 + k_2} [1 - e^{-(k_1+k_2)\tau}]$$

CSTR (solve $C_{i,\text{out}} = C_{i,\text{in}} + \tau R_i$):

$$C_A = \frac{C_{A,0}}{1 + (k_1 + k_2)\tau}$$

$$C_B = \frac{k_1 \tau C_{A,0}}{1 + (k_1 + k_2)\tau}$$

$$C_C = \frac{k_2 \tau C_{A,0}}{1 + (k_1 + k_2)\tau}$$

Both solutions share the same C_B/C_C ratio — see next slide.

Key Result: Selectivity Independent of τ

For equal-order parallel reactions, divide C_B by C_C :

$$S_{B/C} = \frac{C_B}{C_C} = \frac{k_1}{k_2} \quad (\text{constant})$$

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Implications

- $S_{B/C}$ is the **same** in PFR and CSTR
- $S_{B/C}$ does **not** depend on τ
- $S_{B/C}$ does **not** depend on X_A
- Residence time controls conversion only

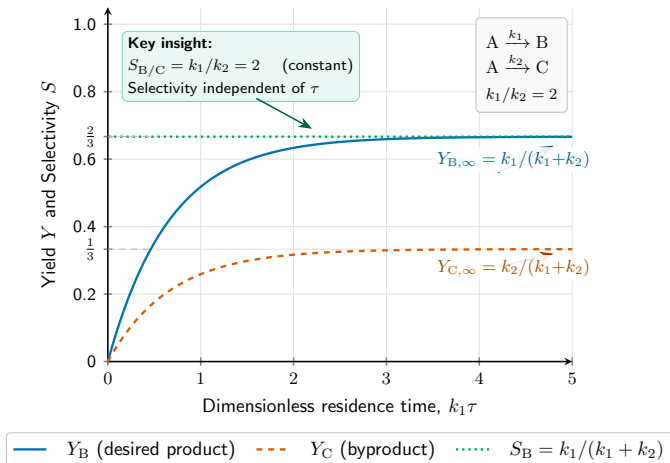
The only lever: temperature

Since k_1 and k_2 depend on T through Arrhenius:

$$\frac{k_1}{k_2} = \frac{A_1}{A_2} e^{-(E_{a,1} - E_{a,2})/RT}$$

Changing T changes the ratio.

Parallel Reactions: Yield & Selectivity vs τ



$k_1/k_2 = 2$: $Y_B \rightarrow 2/3$ and $Y_C \rightarrow 1/3$ at full conversion. $S_{B/C} = 2$ (horizontal green line) — independent of τ .

Non-Equal-Order: Concentration Effect

When reaction orders differ ($r_1 = k_1 C_A^\alpha$, $r_2 = k_2 C_A^\beta$):

$$s_{B/C} = \frac{k_1}{k_2} C_A^{\alpha-\beta}$$

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Reactor selection guide

	Favours B	Reactor
$\alpha > \beta$	High C_A	PFR / Batch
$\alpha < \beta$	Low C_A	CSTR
$\alpha = \beta$	Independent	Either

Physical reasoning

- PFR: high C_A at inlet, decreasing downstream
- CSTR: uniform low C_A everywhere (= exit value)
- Choose the reactor that provides the C_A level favouring B

Temperature Control of Parallel Selectivity

From Arrhenius:

$$\frac{k_1}{k_2} = \frac{A_1}{A_2} \exp\left(-\frac{E_{a,1} - E_{a,2}}{RT}\right)$$

- $E_{a,1} > E_{a,2}$: **raise** T to favour B
- $E_{a,1} < E_{a,2}$: **lower** T to favour B

Crossover temperature (where $k_1 = k_2$):

$$T_{\text{cross}} = \frac{E_{a,1} - E_{a,2}}{R \ln(A_1/A_2)}$$

Design rule

Operate **well away** from T_{cross} on the side that favours B:

If	Then
$E_{a,1} > E_{a,2}$	$T \gg T_{\text{cross}}$
$E_{a,1} < E_{a,2}$	$T \ll T_{\text{cross}}$

Remember

The reaction with the **higher** E_a gains more from raising T .

Worked Example: Parallel Temperature Selection

Problem

$A \rightarrow B$: $E_{a,1} = 80 \text{ kJ/mol}$, $A_1 = 10^{10} \text{ s}^{-1}$. $A \rightarrow C$: $E_{a,2} = 60 \text{ kJ/mol}$, $A_2 = 5 \times 10^7 \text{ s}^{-1}$.
Compare $S_{B/C}$ at 400 K and 500 K. Which temperature favours B?

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$T \text{ (K)}$	$k_1 \text{ (s}^{-1}\text{)}$	$k_2 \text{ (s}^{-1}\text{)}$	$S_{B/C} = k_1/k_2$
400	0.36	0.73	0.49 (C favoured)
500	43.9	27.0	1.63 (B favoured)

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Key insight

Since $E_{a,1} > E_{a,2}$, the desired reaction gains more from raising T . At 400 K, C dominates; at 500 K, B dominates. The crossover temperature is $T_{\text{cross}} = (80-60) \times 10^3 / [8.314 \times \ln(10^{10}/(5 \times 10^7))] = \mathbf{454 \text{ K}}$.

Concept Check — Parallel Selectivity

Concept Check

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 $S_{B/C} = k_1/k_2 = 3$ (independent of X_A and reactor type)
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 $\alpha < \beta$: low C_A favours B $\Rightarrow$ **CSTR** (uniform low C_A)
3. For $A \rightarrow B$ ($E_{a,1} = 50$ kJ/mol) and $A \rightarrow C$ ($E_{a,2} = 80$ kJ/mol), should you raise or lower T to favour B?

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 $E_{a,1} < E_{a,2}$: **lower** T (the undesired reaction gains more from higher T)

From Parallel to Consecutive

What we have so far

- X_A , Y_B , $S_{B/A}$ form a triangle: any two fix the third
- For equal-order parallel reactions, $S_{B/C} = k_1/k_2$ — independent of τ
- Temperature is the **only** lever for parallel selectivity
- For non-equal order, reactor choice (PFR vs CSTR) also matters

The next challenge

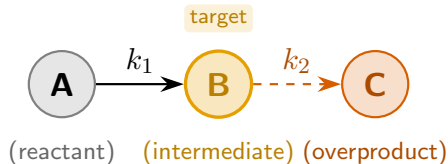
In **consecutive** reactions ($A \rightarrow B \rightarrow C$), the desired product B is an **intermediate** that is consumed by further reaction. There is an optimal τ^* beyond which B is destroyed.

Residence time becomes the critical design variable.

Next: the selectivity window, optimal τ^* , and PFR vs CSTR.

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Consecutive Reactions: $A \rightarrow B \rightarrow C$



First-order kinetics:

$$r_1 = k_1 C_A \quad (\text{A consumed})$$

$$r_2 = k_2 C_B \quad (\text{B consumed})$$

The key difference from parallel

- B is an **intermediate**: formed by r_1 , consumed by r_2
- Net rate of B: $R_B = k_1 C_A - k_2 C_B$
- C_B rises, reaches a **maximum**, then falls
- There is an **optimal** τ^*

Consequence

Operating beyond τ^* **destroys** the desired product. Over-reaction is the enemy.

PFR Solution: Concentration Profiles

Solve $dC_A/d\tau = -k_1 C_A$, then $dC_B/d\tau = k_1 C_A - k_2 C_B$:

$$C_A(\tau) = C_{A,0} e^{-k_1 \tau}$$

$$C_B(\tau) = C_{A,0} \frac{k_1}{k_2 - k_1} (e^{-k_1 \tau} - e^{-k_2 \tau}) \quad (k_1 \neq k_2)$$

$$C_C(\tau) = C_{A,0} - C_A(\tau) - C_B(\tau)$$

Special case $k_1 = k_2 = k$

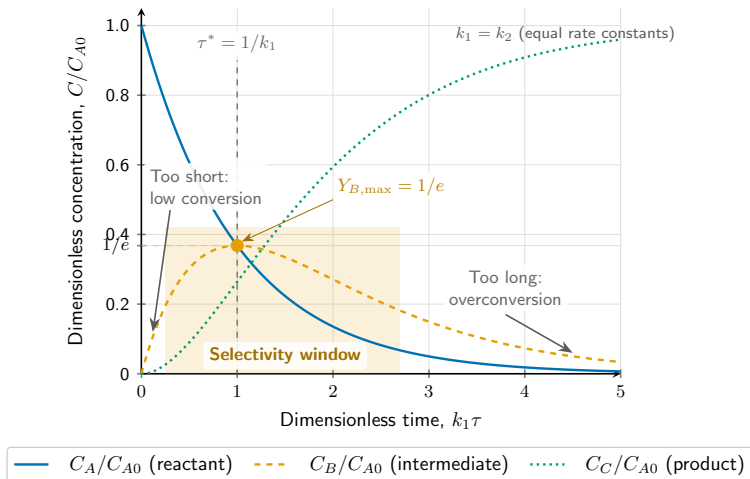
$$C_B = C_{A,0} k \tau e^{-k \tau}$$

(L'Hôpital on the general formula)

Behaviour

- C_A decays exponentially
- C_B rises then falls
- C_C rises monotonically

The Selectivity Window



The shaded region is the **selectivity window**: operate near τ^* to maximise B. Too short $\Rightarrow$ low conversion; too long $\Rightarrow$ B destroyed to form C.

Optimal τ^* for Maximum B

Set $dC_B/d\tau = 0$, i.e. $k_1C_A = k_2C_B$:

$$\tau^* = \frac{\ln(k_2/k_1)}{k_2 - k_1} \quad (k_1 \neq k_2)$$

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Special case $k_1 = k_2 = k$

$$\tau^* = \frac{1}{k}$$

(one time constant of the reaction)

Physical meaning

At τ^* , the rate of B formation **exactly equals** the rate of B consumption.

Before τ^* : formation dominates.

After τ^* : consumption dominates.

Maximum Yield of B

Substitute τ^* back into $C_B(\tau)$:

$$Y_{B,\max} = \left(\frac{k_1}{k_2} \right)^{k_2/(k_2-k_1)}$$

Limiting cases

Condition	$Y_{B,\max}$
$k_1 \gg k_2$	$\rightarrow 1$ (nearly all B)
$k_1 \ll k_2$	$\rightarrow 0$ (B destroyed fast)
$k_1 = k_2$	$= 1/e \approx 0.368$

Design implication

High yield of intermediate requires $k_1/k_2 \gg 1$.

If $k_2 > k_1$ (unfavourable), the maximum yield is inherently low — catalyst development or temperature tuning is needed.

Worked Example: Optimal τ^* in a PFR

Problem

$C_{A,0} = 1.0 \text{ mol/L}$, $k_1 = 0.4 \text{ s}^{-1}$, $k_2 = 1.0 \text{ s}^{-1}$. Find τ^* , $Y_{B,\max}$, and X_A at τ^* .

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$$\tau^* = \frac{\ln(1.0/0.4)}{1.0 - 0.4} = \frac{0.916}{0.6} = \mathbf{1.53 \text{ s}}$$

$$Y_{B,\max} = (0.4/1.0)^{1.0/(1.0-0.4)} = (0.4)^{1.667} = \mathbf{0.217 \text{ (21.7\%)}}$$

$$X_A = 1 - e^{-0.4 \times 1.53} = 1 - e^{-0.612} = \mathbf{0.458 \text{ (45.8\%)}}$$

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Key insight

The low yield (21.7%) reflects $k_2 > k_1$: B is consumed faster than it is formed. Less than half the converted A ends up as B ($S_{B/A} = 0.217/0.458 = 0.47$).

Concept Check — Consecutive Reactions

Concept Check

1. If $k_1 = k_2 = 2.0 \text{ s}^{-1}$, what is τ^* and $Y_{B,\max}$?

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2. What happens to Y_B if we operate at $2\tau^*$? Does Y_B halve, stay the same, or decrease by more than half?

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2. What happens to Y_B if we operate at $2\tau^*$? Does Y_B halve, stay the same, or decrease by more than half? Y_B **decreases** but by **less** than half: at $2\tau^*$, $Y_B = 2e^{-2}/1 = 0.271$ (still 74% of max)

CSTR for Consecutive Reactions

CSTR solution (algebraic):

$$C_A = \frac{C_{A,0}}{1 + k_1\tau}$$

$$C_B = \frac{k_1\tau C_{A,0}}{(1 + k_1\tau)(1 + k_2\tau)}$$

Optimal τ^* (CSTR):

$$\tau_{\text{CSTR}}^* = \frac{1}{\sqrt{k_1 k_2}}$$

Maximum yield (CSTR):

$$Y_{B,\text{max}}^{\text{CSTR}} = \frac{1}{(1 + \sqrt{k_2/k_1})^2}$$

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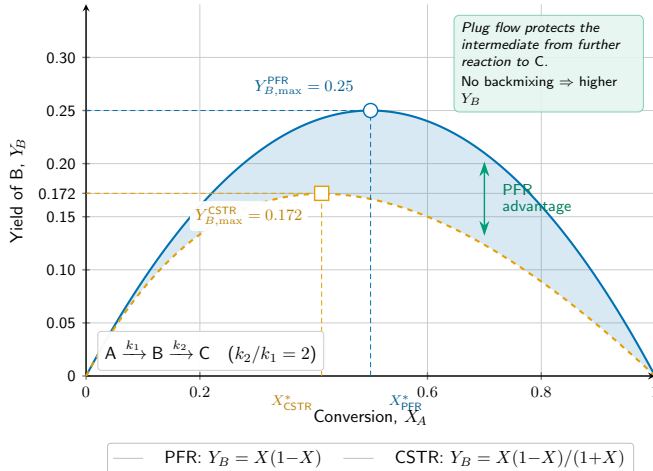
Comparison: $k_1 = k_2$

	PFR	CSTR
τ^*	$1/k$	$1/k$
$Y_{B,\text{max}}$	$1/e$ (36.8%)	$1/4$ (25.0%)

PFR always wins

CSTR backmixing exposes freshly formed B to the full average residence time, causing unnecessary over-reaction.

PFR vs CSTR: Yield of Intermediate



$k_2/k_1 = 2$: PFR achieves $Y_{B,\max} = 0.25$ at $X_A = 0.50$ vs CSTR $Y_{B,\max} = 0.172$ at $X_A = 0.41$. The shaded region shows the PFR advantage.

Worked Example: PFR vs CSTR Comparison

Problem

Same kinetics as before: $k_1 = 0.4 \text{ s}^{-1}$, $k_2 = 1.0 \text{ s}^{-1}$, $C_{A,0} = 1.0 \text{ mol/L}$.

Compare $Y_{B,\max}$ in PFR and CSTR.

Worked Example: PFR vs CSTR Comparison

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Compare $Y_{B,\max}$ in PFR and CSTR.

Reactor	τ^* (s)	$Y_{B,\max}$	X_A	Note
PFR	1.53	0.217 (21.7%)	0.458	Plug flow
CSTR	1.58	0.150 (15.0%)	0.387	Backmixed

Worked Example: PFR vs CSTR Comparison

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CSTR	1.58	0.150 (15.0%)	0.387	Backmixed

Key insight

PFR yield is **45% higher** than CSTR ($0.217/0.150 = 1.45$). Backmixing in the CSTR destroys intermediate B. For consecutive reactions, **always prefer PFR** (or approach plug flow with CSTRs in series).

Multiple CSTRs Approach PFR Performance

CSTRs in series reduce backmixing:

Configuration	$Y_{B,\max}$
1 CSTR	0.150
3 CSTRs	0.190
10 CSTRs	0.210
∞ CSTRs = PFR	0.217

($k_1 = 0.4$, $k_2 = 1.0 \text{ s}^{-1}$; equal total τ)

Why it works

Each CSTR sees a **smaller** concentration jump than a single CSTR. As $N \rightarrow \infty$, the cascade becomes a differential plug-flow element.

Practical guideline

3–5 CSTRs in series capture $\sim 90\%$ of the PFR advantage. Beyond that, diminishing returns vs added complexity.

Temperature Effects on Consecutive Selectivity

Temperature affects both the k_1/k_2 ratio **and** $\tau^*(T)$:

- $E_{a,1} > E_{a,2}$: raising T increases $k_1/k_2 \Rightarrow$ **improves** $Y_{B,\max}$
- $E_{a,1} < E_{a,2}$: raising T decreases $k_1/k_2 \Rightarrow$ **reduces** $Y_{B,\max}$

Same Arrhenius rule

The reaction with the higher E_a wins at higher T — applies to **both** parallel and consecutive.

Temperature programming

For $E_{a,1} > E_{a,2}$:

- Start **hot** (fast $A \rightarrow B$)
- Cool as B accumulates (slow $B \rightarrow C$)
- Can exceed isothermal $Y_{B,\max}$

Applicable to PFR and batch, not CSTR (uniform T).

Trade-off

Lower $T \Rightarrow$ better S but longer τ^*
 $\Rightarrow$ larger reactor volume needed.

Worked Example: Temperature Effect on $Y_{B,\max}$

Problem

$A \rightarrow B$: $E_{a,1} = 50 \text{ kJ/mol}$, $A_1 = 10^6 \text{ s}^{-1}$. $B \rightarrow C$: $E_{a,2} = 70 \text{ kJ/mol}$, $A_2 = 10^8 \text{ s}^{-1}$.
Compare τ^* and $Y_{B,\max}$ at 400 K and 450 K.

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$T \text{ (K)}$	$k_1 \text{ (s}^{-1}\text{)}$	$k_2 \text{ (s}^{-1}\text{)}$	$\tau^* \text{ (s)}$	$Y_{B,\max}$	Observation
400	0.295	0.0722	6.31	63.4%	Higher yield
450	1.57	0.749	0.90	50.9%	Lower yield

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Key insight

Since $E_{a,2} > E_{a,1}$, raising T accelerates $B \rightarrow C$ **more** than $A \rightarrow B$. Lower temperature gives 12.5 percentage points more yield, but requires $7\times$ longer residence time.

Concept Check

1. For $A \rightarrow B \rightarrow C$ with $E_{a,1} = 100$ and $E_{a,2} = 60$ kJ/mol, should you raise or lower T to maximise Y_B ?

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3. A parallel process has $S_{B/C} = 0.8$ at 400 K and $S_{B/C} = 1.5$ at 500 K. Which reaction has the higher E_a ? $S_{B/C}$ increases with $T \Rightarrow$ the desired reaction (B) has the **higher** E_a

Industrial Selectivity Strategies

Ethylene oxide (parallel)

- $\text{C}_2\text{H}_4 + \frac{1}{2}\text{O}_2 \rightarrow \text{EO (desired)}$
- $\text{C}_2\text{H}_4 + 3\text{O}_2 \rightarrow 2\text{CO}_2 + 2\text{H}_2\text{O}$
- Ag/ α -Al₂O₃ catalyst with Cl promoter
- Per-pass $X \approx 8\text{--}12\%$
- Recycle gives $>95\%$ overall yield

Strategy: low conversion + recycle; $T \approx 250^\circ\text{C}$; Cl inhibits total oxidation.

Toluene chlorination (consecutive)

- Toluene $\rightarrow$ benzyl chloride $\rightarrow$ benzal chloride
- Target: monochlorinated product
- Use PFR with precise τ control
- Excess toluene/Cl₂ ratio > 1 suppresses over-chlorination

Strategy: operate at τ^* ; high substrate/reagent ratio favours mono-substitution.

Both strategies exploit the principles developed today: parallel $\rightarrow$ temperature & conversion control;
consecutive $\rightarrow \tau$ control.

Selectivity Engineering: A Decision Framework

Question	Parallel	Consecutive	Tool
What controls S ?	k_1/k_2 ratio	τ (selectivity window)	Kinetics
Reactor choice?	PFR = CSTR for S (equal order)	PFR > CSTR always	Design
T effect?	Higher E_a wins at higher T	Same rule	Arrhenius
Industrial fix?	Low X + recycle	Operate at τ^*	Process

The selectivity engineer's toolbox

Network topology $\xrightarrow{\text{kinetics}}$ Rate ratio $\xrightarrow{\text{Arrhenius}}$ $T, \tau, \text{reactor}$ $\xrightarrow{\text{design}}$ Maximise Y_B

Choose the right **lever** for each network: temperature for parallel, residence time for consecutive, reactor type when orders differ.

- 1 Part I: Definitions & Parallel Reactions
- 2 Part II: Consecutive Reactions & Design
- 3 Summary & Self-Study**

Key Equations Card

Definitions & Parallel:

1. **Conversion:** $X_A = (C_{A,0} - C_A)/C_{A,0}$
2. **Yield:** $Y_B = C_B/C_{A,0}$
3. **Selectivity:** $S_{B/A} = Y_B/X_A$
4. **Parallel S:** $S_{B/C} = k_1/k_2$ (equal order)
5. T_{cross} :
 $T_{\text{cross}} = (E_{a,1} - E_{a,2})/[R \ln(A_1/A_2)]$

Consecutive & Design:

6. C_B (PFR): $C_B = C_{A,0} \frac{k_1}{k_2 - k_1} (e^{-k_1 \tau} - e^{-k_2 \tau})$
7. τ^* (PFR): $\tau^* = \ln(k_2/k_1)/(k_2 - k_1)$
8. $Y_{B,\text{max}}$ (PFR): $Y_{B,\text{max}} = (k_1/k_2)^{k_2/(k_2 - k_1)}$
9. τ^* (CSTR): $\tau_{\text{CSTR}}^* = 1/\sqrt{k_1 k_2}$
10. $Y_{B,\text{max}}$ (CSTR): $Y_{B,\text{max}}^{\text{CSTR}} = 1/(1 + \sqrt{k_2/k_1})^2$

Ten equations — from definitions through optimal design for both reactor types.

NB5: Self-Study Exercises

Exercises (complete before Week 5)

Ex	Topic	Priority
1	Parallel Y_B , $S_{B/C}$ vs τ	Essential
2	Consecutive profiles & τ^*	Essential
3	PFR vs CSTR comparison	Essential
4	T optimisation of $Y_{B,\max}$	Essential
5	CSTR cascade convergence	Bonus

26 cells | $\sim 2-3$ h |

<https://github.com/ExaPsi/2302337>

Self-study from Chapter 5

Topics not covered in today's lecture:

- Recycle reactor yield derivation (§5.2)
- Mixed networks (parallel + consecutive) (§5.4)
- Non-equal-order consecutive kinetics (§5.5)
- Temperature programming in batch (§5.6)

Notebook roadmap

Week 1–2	NB1, NB2, NB3
Week 3	NB4
Week 4	NB5
Week 5	NB6

Exam details

- **Date & time:** Sunday 20 April 2026, 08:00–10:00 (120 min)
- **Format:** Jupyter notebook + written answer sheet (100 points)
- **Coverage:** Chapters 1–5 (Adsorption, Kinetics, Reactors, Transport, Selectivity)
- **Open book**, internet allowed, AI tools allowed

4 Problems

#	Topic	Pts
1	BET surface area	22
2	Arrhenius/Eyring	28
3	Transport limitations	24
4	Selectivity engineering	26

Preparation

- Complete NB2–NB5 exercises
- Review key equations (Thiele, η , τ^* , $Y_{B,\max}$)
- Practice Python: regression, plotting, `np.cosh/np.log`

Good luck on the exam!